A-A-54165
12 April 1991

COMMERCIAL ITEM DESCRIPTION

BUR, BONE, Tungsten Carbide
GENERAL SPECIFICATION FOR

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-B-42017 which is cancelled.

This commercial item description covers the general requirements for tungsten carbide, bone burs used in DoD Deployable Medical Systems (DMS) and is intended to be used in conjunction with the commercial item description sheets or procurement documents covering any additional requirements applicable to each individual bur.

Salient characteristics:

Classification. The surgical burs shall be of the following head shapes and shank lengths as specified in the applicable commercial item description sheets.

Shank:	Shank length:
a. Round Cutting	
b. Side Cutting, Straight	1. Medium
c. Side Cutting, Tapered	2. Long
d. Oval Cutting	
e. Barrel Cutting	

Material. The burs shall be fabricated of tungsten carbide. In the absence of certificates of quality, for the raw material, the samples selected for analysis shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute Method 111 shall govern.

Style, design and dimensions. The burs shall be tungsten carbide burs of one-piece construction, suitable for use during bone surgery when used in a pneumatic bone drill. The bur head shall have properly and uniformly sharpened cutting edges. The style, design, and dimensions of the burs shall be as specified in the applicable commercial item description sheet.

Runout. The total indicated runout of the burs shall not exceed 0.002 inch.

AMSC N/A FSC 6515

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Finish. The bur heads shall have a finely honed finish with sharpened blades. The neck and shank of the burs shall be smooth, and any safeline present on the shank shall have an etched finish.

Autoclavability. The burs shall show no evidence of corrosion after being cleaned in accordance with manufacturer instructions, and then subjected to 5 autoclave cycles of 270 ± 5 °F, and 27 ± 1 psi for a minimum of 20 minutes each.

Workmanship. The burs shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each. (EA) One bur of the size and shape specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, at inspection level II, with an AQL of 1.5 (percent defective), except all dimensions other than head diameter shall be at inspection level S-1, with an AQL of 2.5 (percent defective) and workmanship finish, and instruction shall be at inspection level II with an AQL of 4.0 (percent defective). The unit of product for examination purposes shall be one bur. Where feasible, the same burs shall be used for the determination of two or more examination characteristics.

Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturers gages may be used if properly calibrated checking standards are used in conjunction with the gage.

Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadowgraph or other suitable equipment of equal ac curacy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360° cycle rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Visual examination. Examination for possible bur defects not perceptible to the unaided eye shall be made using a 30x magnification, unless otherwise specified in the individual Commercial Item Description sheet.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, at inspection level S-1, with an AQL of 1.5 (percent defective). The unit of product for test purposes shall be one bur. Where feasible, the same burs shall be used for the determination of two or more test characteristics.

Tests. Tests shall be conducted to determine compliance with this document.

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Contract certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulation promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contact and/or order.

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NOTES.

Ordering Data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable specification sheet which shall include this specification letter and number, and the slash number. Examples:

A-A-54165/1	Bur, Bone, Tungsten Carbide, Side Cutting, 19 mm Cutting Length, Tapered, Long Shank
A-A-54165/2	Bur, Bone, Tungsten Carbide, Round Cutting, 1.5 mm Head Diameter,Long Shank
A-A-54165/3	Bur, Bone, Tungsten Carbide, Round Cutting, 2 mm Head Diameter, Long Shank
A-A-54165/4	Bur, Bone, Tungsten Carbide, Round Cutting, 2 mm Head Diameter, Medium Shank
A-A-54165/5	Bur, Bone, Tungsten Carbide, Side Cutting, 7.5 mm Cutting Length, Tapered, Medium Shank
A-A-54165/6	Bur, Bone, Tungsten Carbide, Side Cutting, 19 mm Cutting Length, Straight, Medium Shank
A-A-54165/7	Bur, Bone, Tungsten Carbide, Round Cutting, 1.5 mm Head Diameter, Medium Shank
A-A-54165/8	Bur, Bone, Tungsten Carbide, Side Cutting, 7.5 mm Cutting Length, Tapered Medium Shank
A-A-54165/9	Bur, Bone, Tungsten Carbide, Oval Cutting, 8 mm Cutting Length, Medium Shank
A-A-54165/10	Bur, Bone, Tungsten Carbide, Oval Cutting, 10 mm Cutting Length, Medium Shank
A-A-54165/11	Bur, Bone, Tungsten Carbide, Oval Cutting, 8 mm Cutting Length, Long Shank
A-A-54165/12	Bur, Bone, Tungsten Carbide, Oval Cutting, 10 mm Cutting Length, Long Shank
A-A-54165/13	Bur, Bone, Tungsten Carbide, Side Cutting, 5 mm Cutting Length, Straight Medium Shank
A-A-54165/14	Bur, Bone, Tungsten Carbide, Barrel Cutting, Medium Shank
A-A-54165/15	Bur, Bone, Tungsten Carbide, Barrel Cutting, Long Shank
A-A-54165/16	Bur, Bone, Tungsten Carbide, Round Cutting, 3 mm Head Diameter, Long Shank
A-A-54165/17	Bur, Bone, Tungsten Carbide, Round Cutting, 5 mm Head Diameter, Medium Shank
A-A-54165/18	Bur, Bone, Tungsten Carbide, Round Cutting, 5 mm Head Diameter, Long Shank
A-A-54165/19	Bur, Bone, Tungsten Carbide, Round Cutting, 4 mm Head Diameter, Long Shank

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A-A-54165/20	Bur, Bone, Tungsten Carbide, Round Cutting, 4 mm Head Diameter, Medium Shank
A-A-54165/21	Bur, Bone, Tungsten Carbide, Side Cutting, 4.5 mm Cutting Length, Straight, Medium Shank
A-A-54165/22	Bur, Bone, Tungsten Carbide, Side Cutting, 10 mm Cutting Length, Tapered, Medium Shank
A-A-54165/23	Bur, Bone, Tungsten Carbide, Side Cutting, 12 mm Cutting Length, Tapered, Long Shank

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4877
Location: WORD/BURBONET

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54165	2. DOCUMENT DATE (YYMMDD) 12 April 1991

3. DOCUMENT TITLE
BUR, BONE, TUNGSTEN CARBIDE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Personnel Support Center	b. TELEPHONE (Include Area Code) (1) Commercial: (215)737-2870 (2) AUTOVON: 444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSTE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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